UNDER REVIEW

RESCIENCE C

Replication / Computational Biology

[Re] Paradoxical dependencies of tumor dormancy and progression on basic cell kinetics

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Introduction

Enderling et al. [1] introduced a cellular automaton of cancer stem cells, which divide without limit, and their non-stem progeny, which can divide only a limited number of times. Their central and counter-intuitive result is that increasing the spontaneous death rate of non-stem cells can make tumours grow larger in the long run: dying non-stem cells free space in which stem cells can divide, so the stem cell pool, and eventually the tumour, expands. Poleszczuk and Enderling [2] later gave a compact specification of the model with explicit parameter values and a reference tumour size. We re-implemented the model from this specification and compared it with both papers. The authors' code, announced in [2], could not be found.

Methods

Model. As specified in [2]: a two-dimensional square lattice with one cell per $10\,\mu\text{m}$ site and Moore neighbourhood. Stem cells are immortal and divide without limit; at division they produce another stem cell with probability $p_s = 0.1$, otherwise a non-stem cell with proliferation capacity $\rho_{\max} = 10$. A non-stem cell divides at most $\rho_{\max}$ times; its capacity decreases at each division and is inherited by the daughter. A cell needs a free neighbouring site to divide or move; cells without one are quiescent. At each hourly step, the cells present at the start of the step are updated once, in random order: with probability $p_d = 1/24$ (cell cycle 24 h) a cell attempts to divide, otherwise it migrates with probability p_m to a free neighbour chosen uniformly. One stem cell starts in the centre; tumours grow for 180 days.

Spontaneous death. A non-stem cell dies with probability α at a division attempt, or when its capacity is exhausted. [2] does not say when death is tested; we took this rule from the later methods chapter of the same group [3].

Migration. [2] gives the motility as $\mu = 100\,\mu\text{m}$ per day and $p_m = \mu\,\Delta t$. Read in cell widths (10 cells per day, 24 steps per day), $p_m = 10/24$; read literally in micrometres, p_m exceeds one, i.e. a cell that does not divide always attempts to move ($p_m = 1$). We report both readings.

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Code is available at https://github.com/quantumproteinsai/rescience-bozic2013-enderling2009 – DOI 10.5281/zenodo.23437598..

Implementation and ensembles. C++ with one tumour per OpenMP thread, fixed lattices of 1000×1000 sites (1400×1400 for the migration sweep, whose largest tumours are bigger) that no tumour reached the edge of, and the 64-bit Mersenne Twister of the C++ standard library seeded per tumour. 32 tumours per death rate ($\alpha = 0$, 0.01, 0.05, 0.1, 0.2 and 0.3) with $p_m = 10/24$; 16 tumours per migration probability. Means are given with standard errors.

Results

The paradox is reproduced. Tumours grow larger as the death rate of non-stem cells increases from 0 to 0.2: 56,922 cells at $\alpha = 0$ against 167,773 at $\alpha = 0.2$ after 180 days, with 12-fold more stem cells (Table 1, Figure 1), as reported in [1]. More than 70 % of cells have no free neighbour (71 % at $\alpha = 0.01$), as reported in [2].

Table 1. Tumours after 180 days ($p_m = 10/24$, 32 tumours each; mean ± standard error).

α	cells	stem cells	without free neighbour
0	56,922 ± 3,068	150 ± 9	70.2 %
0.01	60,692 ± 2,604	173 ± 7	71.3 %
0.05	85,645 ± 3,529	305 ± 13	75.6 %
0.1	116,699 ± 3,967	571 ± 19	78.1 %
0.2	167,773 ± 4,932	1811 ± 53	83.1 %
0.3	179,227 ± 4,500	4794 ± 122	84.7 %

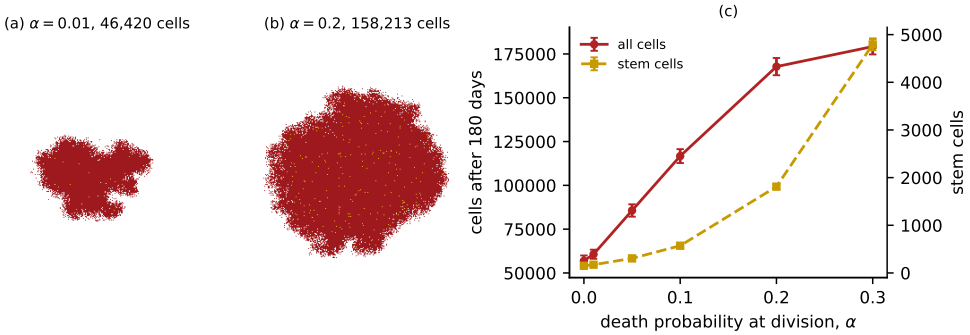


Figure 1. Tumours after 180 days. (a, b) One tumour at $\alpha = 0.01$ and $\alpha = 0.2$ (same seed; stem cells in yellow; each panel 7 mm wide). (c) Final numbers of cells and stem cells against α (mean ± s.e., 32 tumours each).

The published tumour size is not reproduced. [2] reports about 140,000 cells after 180 days at $\alpha = 0.01$. With $p_m = 10/24$ we obtain 60,692 cells (mean of 32 tumours), with $p_m = 1$ 328,155 (16 tumours). The final size grows steeply with p_m, and the published size corresponds to $p_m \approx 0.61$ in our implementation (Figure 2).

Discussion

The qualitative result of [1], the paradoxical growth of tumours with a higher non-stem cell death rate, is robust: it follows from an independent implementation of the published specification. The quantitative tumour size is not reproduced, because the specification leaves open two rules that strongly affect it: the migration probability per step

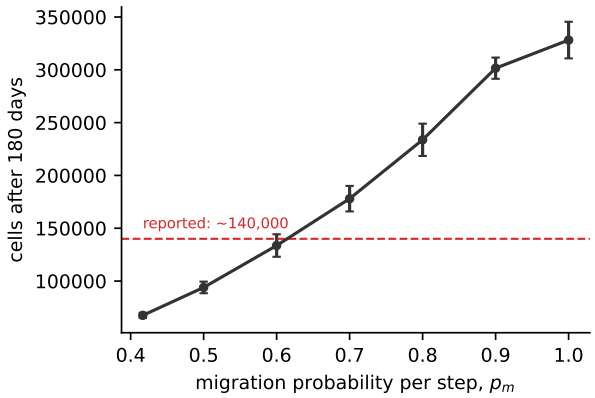


Figure 2. Tumour size after 180 days against the migration probability per step ($\alpha = 0.01$, 16 tumours each, mean ± s.e.). Dashed: about 140,000 cells reported in [2]. The two readings of $\mu = 100$ µm per day are $p_m = 10/24$ and $p_m = 1$.

and the moment at which spontaneous death is tested. Between the two readings of the motility, the final size changes more than five-fold. Other implementations read the same specification differently: the sequential C++ code of a group that parallelised the model [4, 5] takes $\mu\,\Delta t = 100/24$ literally and normalises it with the death and division probabilities, so that a cell with a free neighbour always dies, migrates or divides, and it tests spontaneous death at every step rather than at a division attempt, with no death when the proliferation capacity is exhausted. Quantitative uses of this model should state these rules explicitly; ours are given above and in the code. We contacted the corresponding author about the migration rule and the original code.

Code, data and reproducibility

Code: https://github.com/quantumproteinsai/rescience-bozic2013-enderling2009, archived at doi:10.5281/zenodo.23437598. The model is `src/ca/ca_reference.cpp`; `make all` rebuilds it and regenerates every number, table and figure from fixed seeds. For reviewers, `make quick-check` reproduces 11 representative published results exactly, from scratch, in about two minutes on a laptop. The exact numbers are reproduced with GCC and its standard library; other C++ standard libraries implement the random distributions differently and give statistically equivalent, not identical, results.

Use of AI assistance

The software, simulations, analyses and the first draft of this article were produced with an AI coding assistant (Claude Opus 5.5, Anthropic, used through Claude Code), under the direction of the author, who defined the study, chose the comparisons, checked the results against the original publications and revised the text. The AI is not an author; the author takes full responsibility for the content.

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